

Lab 10: Comparison of Deep Learning Frameworks and Architectures

Course: Programming for Data Science
Professor: Dr. Carlos J. Costa

April 12, 2026

Introduction

In this lab, you will analyze and compare different deep learning frameworks and model architectures using the notebooks *CompareArchitecture.ipynb* and *CompareFrameworks.ipynb*. The goal is to understand how implementation choices affect model definition, training, and results.

You are expected to:

- Understand and compare how different frameworks define and train neural networks,
- Analyze the impact of using the same model across different frameworks,
- Evaluate how different architectures affect performance under the same framework,
- Identify possible improvements in the provided implementations.

Tasks

1. Framework syntax and workflow

- Explain the syntax used in each framework.
- Describe how the model architecture is defined.
- Specify what data preparation steps are required.
- Explain how models are trained (e.g., training loop, compilation, fitting process).

2. Differences between frameworks

- Why do results differ when using the same model in different frameworks?

3. Differences between architectures

- Why do results differ when using different architectures with the same framework?

4. Possible improvements

- Identify improvements in the code.
- Consider data preprocessing, model design, training strategy, evaluation, and code structure.